

Krishnaa Vadivel | Computational Physics | Distributed Computing | Machine Learning

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Summary

Computational physics researcher building distributed scientific software and using machine learning to accelerate calculations for first-principles materials research. Experience spans Python/C++, CUDA/ROCm, MPI/OpenMP, Linux-based HPC environments, and Docker containers for packing applications across environments.

Research Experience

Yale University, Electrical Engineering

PhD Research Assistant

2021–present

- Developed distributed simulation software in Python and C++ for first-principles workloads as part of an INCITE award using exascale resources, with MPI, OpenMP, CUDA, ROCm, PETSc, and SLEPc on Frontier at OLCF, Perlmutter at LBNL, and related national-lab clusters.
- Formulated coherent-state and path-integral viewpoints for exciton-phonon dynamics in self-trapped excitons and excitonic polarons, connecting DFT, DFPT, and GW/BSE-style methods.
- Used PyTorch, PyTorch Geometric, and equivariant graph neural networks to accelerate self-trapped-exciton calculations and related materials simulations.
- Packaged scientific applications in Docker containers for deployment across different computing environments and used distributed debuggers such as TotalView and Linaro DDT to track down bugs on HPC systems.
- Co-authored a 2025 *Journal of the American Chemical Society* paper on ultrafast self-trapped exciton formation in lead-free double halide perovskites and presented related APS research in 2024, 2025, and 2026.

Industry Experience

Probe Technologies Inc. / Weatherford Wireline Services

Software and Embedded Systems Engineer

2019–2021

- Developed CUDA-based software for high-volume acoustic waveform processing, visualization, and field-tool diagnostics in geological well logging systems.
- Built full-stack server applications with C#, ASP.NET, and Microsoft SQL Server for database design, real-time remote tool testing, and calibration of temperature and pressure sensors.
- Applied finite-element analysis with dolfinx for frequency analysis of steel pipes in well environments to support physics-driven engineering studies.

FindLight

Photonics Technical Writer

2018–2019

- Wrote clear technical content on lasers, optics, and photonics devices for engineering audiences and product-focused readers.

Selected Projects

Exciton-Phonon HPC Research: Developed distributed first-principles calculations for self-trapped excitons, many-body theory, and scalable materials simulations.

Equivariant Scientific ML: Developed PyTorch Geometric equivariant models to accelerate molecular and excitonic calculations.

Meep Ring Resonator FDTD: Designed a nanophotonic ring-resonator band-pass filter in telecom bands using

Meep-based FDTD simulation.

DOLFINx Finite-Element Analysis: Built 2D DOLFINx/PETSc simulations for transient heat diffusion, Poisson potential around a circular cutout, and incompressible flow past an immersed cylinder, with PyVista and Matplotlib results.

Education

Yale University

PhD, Electrical Engineering

2021–present

Yale University

MPhil, Electrical Engineering

2023

Yale University

MS, Electrical Engineering

2023

Cornell University

MEng, Electrical and Computer Engineering

2017–2018

Cornell University

BS, Electrical and Computer Engineering

2013–2017

Technical Skills

Programming: Python, C, C++, CUDA, JavaScript, Bash, SQL, C#

Scientific Computing: MPI, OpenMP, PETSc, SLEPc, NumPy, SciPy, SymPy, matplotlib, dolfinx

Machine Learning: PyTorch, PyTorch Geometric, scikit-learn, equivariant graph neural networks

Platforms: Linux, macOS, Windows, Docker, Docker Compose, Kubernetes, gcloud, Frontier, Perlmutter

Domains: First-principles materials modeling, many-body theory, distributed computing, computational physics, numerical PDEs